Vector Spaces Worksheet

Determine whether or not the given set of objects, with scalar multiplication and vector addition defined in the most obvious way unless otherwise specified, represents a vector space. If it does, give a basis. If it is not, give at least one property it fails to satisfy.

1) The set of ordered triples (x, y, z) of real numbers satisfying $2x + 3y - z = 5$.

2) The set of ordered triples (x, y, z) of real numbers satisfying $2x + 3y - z = 0$.

3) The set of ordered triples (x, y, z) of real numbers satisfying $\begin{matrix} 2x + y - 3z = 0 \\ 2y + 5z = 0 \end{matrix}$.

4) The set of ordered triples (x, y, z) of real numbers satisfying $\begin{matrix} 2x + y - z = 4 \\ 2y + 5z = 0 \end{matrix}$.

5) The set of ordered triples (x, y, z) of real numbers satisfying $2y + 5z = 0$.

6) The set of ordered pairs (x, y) for which $x = a + 3b$ and $y = 2a - b$ with $a, b \in \mathbb{R}$.

7) The set of ordered pairs (x, y) of real numbers satisfying $x^2 - 4x + y^2 = 0$.

8) The set of ordered pairs (x, y) of real numbers satisfying $x^2 + y^2 = 0$.

9) The set of polynomials $p(x)$ of degree 4.

10) The set of polynomials $p(x)$ of degree less than or equal to 6.

11) The set of polynomials $p(x)$ of degree less than or equal to 4 satisfying $p(3) = 0$.

12) The set of polynomials $p(x)$ of degree less than or equal to 4 satisfying $p(0) = 3$.

13) The set of polynomials $p(x)$ for degree less than or equal to 4 satisfying $p(-1) = p(2)$.

14) The set of polynomials $p(x)$ for degree less than or equal to 4 satisfying $p(-1) = p(2)$ and $p(3) = 0$.

15) The set of polynomials $p(x)$ for degree less than or equal to 4 satisfying $p(-1) = p(2) + 1$ and $p(3) = 0$.

16) The set of vectors $\begin{bmatrix} 2a-b+3 \\ a+2b-1 \\ 3a+2b+1 \end{bmatrix}$, with $a, b \in \mathbb{R}$.

17) The set of vectors $\begin{bmatrix} 2a-b+3c \\ a+2b-c \\ 3a+2b+5c \end{bmatrix}$, with $a, b, c \in \mathbb{R}$.

18) The set of vectors $\begin{bmatrix} 2a-b+3c \\ a+2b-c \\ 3a+2b+5c \end{bmatrix}$, with $a, b, c \in \mathbb{R}$ and $2a+b-5c = 0$.

19) The set of vectors $\begin{bmatrix} 2a-b+3c \\ a+2b-c \\ 3a+2b+5c \end{bmatrix}$, with $a, b, c \in \mathbb{R}$ and $2a+b-5c = 3$.

20) The set of vectors $\begin{bmatrix} 2a-b+3 \\ a+2b-1 \\ 3a+2b+2 \end{bmatrix}$, with $a, b \in \mathbb{R}$.